Why medication safety needs a spatial interface

1 Research area

Augmented reality for patient-facing medication safety: joining real-time computer vision, drug knowledge graphs and in-situ 3D presentation on a commodity smartphone.

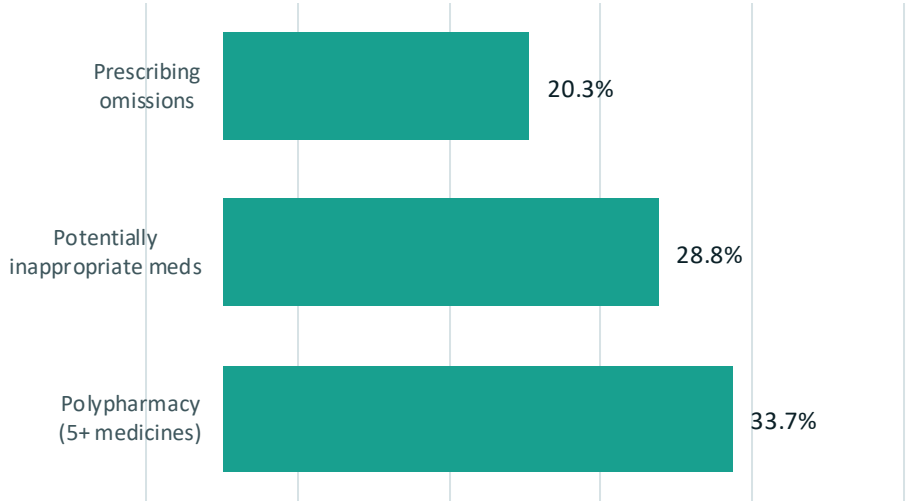
2 Why it matters

Medication errors are among the most preventable causes of patient harm. The information needed to avoid them already exists on the box — but it is small, dense, and written for pharmacists, not patients.

3 Current challenges

Brand names hide shared active ingredients · leaflets are unreadable for low-vision and elderly users · interaction checking requires a separate app and typing every drug name · existing AR tools handle one product at a time.

Unsafe medication use among older adults in India



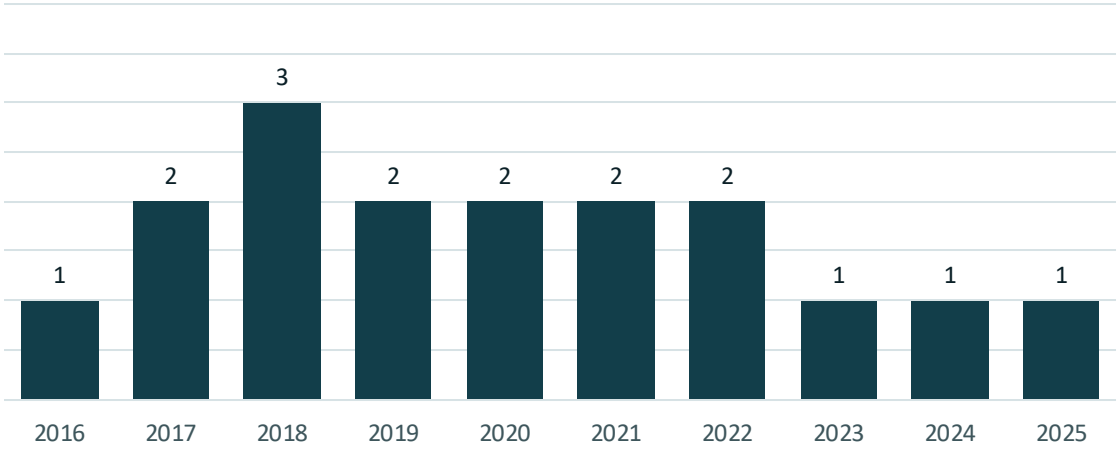
33.7% of 600 older adults surveyed across six Indian cities were on five or more medicines concurrently; 28.8% of prescriptions contained a potentially inappropriate medication.

Source: cross-sectional study of polypharmacy and self-medication among older adults in Indian urban communities, Scientific Reports, 2025.

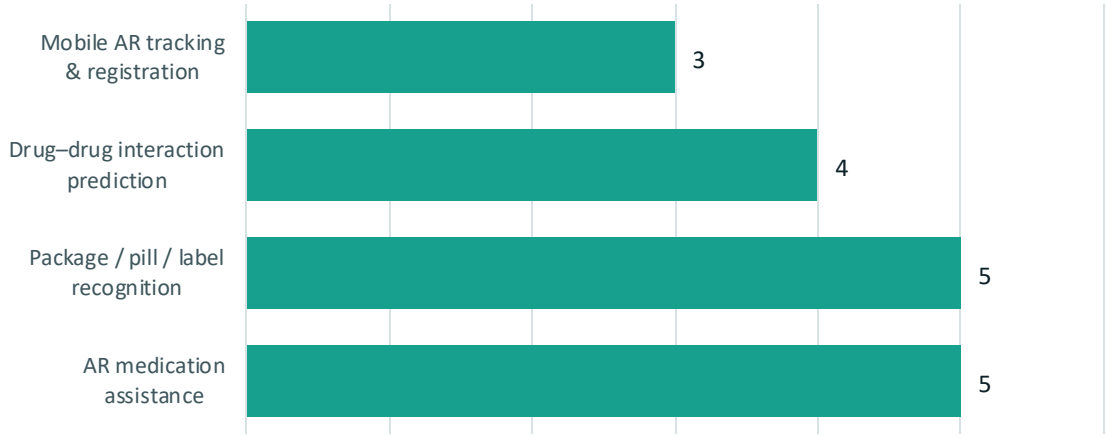
Scope of the review: 17 papers across four strands

Papers were selected from IEEE Xplore, Springer, Elsevier, ACM, PNAS and PMC-indexed journals, prioritising work from 2016 onward that either (a) places medication information in an AR view, (b) recognises medicine packaging or pills, (c) predicts drug–drug interactions, or (d) addresses the tracking and registration problem that any mobile AR system must solve.

Publication year



Research strand



What the strands never meet on: perception work stops at identifying one object, interaction work stops at a server-side prediction, and AR work stops at tracking a generic object. No reviewed paper closes the loop from packaging → active ingredient → interaction reasoning → registered overlay.

Strand A — AR-based medication assistance

Paper / Authors / Venue	Methodology	Dataset & tools	Major findings and advantages	Limitations
Medicine Drug Name Detection Based Object Recognition Using AR Ch. Rupa et al. · 2022 Frontiers in Public Health, 10:881701	OCR on pack text → NLP → generic name via a drug-name-recognition database; ARCore renders a label and 3D model over the pack.	In-house DNR database; ARCore, Android SurfaceView.	Closest prior work. A full markerless AR identification loop is achievable on a commodity phone, and OCR on packaging is a viable route.	One medicine per session. No dosage, timing, interaction or duplicate checking. Small closed database; no standard accuracy metric.
Smart AR Assisting Patients with Medication Pillbox Self-Management M. Blusi, J. C. Nieves · 2019 MedInfo, Stud. Health Technol. Inform., 264:521–525	A Medication Coach Intelligent Agent on HoloLens guides the user through filling a weekly pillbox; questionnaire evaluation.	HoloLens 1; real regimens; 15 participants aged 17–76.	Hands-free spatial guidance for polypharmacy rated feasible and acceptable — the human-factors evidence that AR guidance is welcomed.	Headset cost blocks deployment; willingness fell with age. n = 15, no control. No interaction or duplicate reasoning.
Microsoft HoloLens — A mHealth Solution for Medication Adherence M. Ingeson, M. Blusi, J. C. Nieves · 2018 AI in Health (AIH @ IJCAI), pp. 99–115	Agent-based mixed-reality architecture: a BDI-style agent issues contextual reminders and dialogue in the headset.	HoloLens prototype; simulated medication regimens.	Establishes the pattern of an autonomous agent inside the AR view rather than a passive overlay — precedent for our reasoning module.	Concept prototype; no quantitative accuracy reported. Expensive headset; no safety checking beyond reminders.
AR Technology: A Systematic Review on Gaming Strategy for Medication Adherence R. O. Adetunji et al. · 2022 Studies in Computational Intelligence, 1061, Springer	Systematised scoping review of IoT, AR and gamification components for medication adherence.	Scholar, ACM, ScienceDirect, IEEE Xplore, SpringerLink.	Consolidates the AR-for-adherence design space and confirms adherence, not safety, dominates the field — justifying our shift to checking.	Review only; no system, dataset or metrics. Gamification framing sidesteps interaction and duplication risk.
Healthcare System for Elders with Automatic Drug Label Detection P. Tangtisanon · 2016 ICCAS, pp. 666–670	Classical pipeline: segment the label region, then template and text matching to identify the drug.	Custom label image set; embedded camera.	Early proof that automated label reading can aid elderly medication handling at very low compute; a non-deep-learning baseline.	Not AR — 2D screen readout. Rigid templates fail on glare, rotation and repackaging. No clinical reasoning.

All five works identify or narrate a single medicine. None reason about relationships between two or more physical packages in the same scene.

Strand B — Package, pill and label recognition

Paper / Authors / Venue	Methodology	Dataset & tools	Major findings and advantages	Limitations
MobileDeepPill: Small-Footprint Mobile Deep Learning for Unconstrained Pill Images X. Zeng, K. Cao, M. Zhang · 2017 ACM MobiSys	Three parallel CNNs over colour, grey and gradient views, compressed by knowledge distillation to run on the phone.	NIH NLM Pill Image Recognition Challenge.	Landmark proof that clinically useful pill recognition fits a mobile footprint — direct support for our on-device design.	Loose pills only, never packaging. Static capture; no AR registration, dosage or interaction reasoning.
Pharmaceutical Blister Package Identification Based on Induced Deep Learning Y. Han et al. · 2021 IEEE Access	Two-stage induced deep learning: human packaging heuristics steer image adjustment before a CNN classifier.	Custom blister-package database; YOLOv2, ResNet, SENet baselines.	Reported 99.98% vs 86.5–96.2% F1 for baselines, and robust on near-identical packs — the hardest duplicate-detection case.	Closed-set, fixed capture, one pack per image. No text understanding, so it cannot generalise to unseen brands or read dosage.
DLI-IT: Drug Label Identification through Image and Text Embedding X. Liu et al. · 2020 BMC Med. Inform. Decis. Mak., 20(1):68	Learns a joint embedding over the label image and its extracted text, then retrieves the matching record.	FDA drug label image + text corpus.	Image + text fusion clearly beats image-only matching on text-dense labels — the basis for our fused OCR-plus-appearance resolver.	Server-side batch retrieval, not real-time. No AR, no dose or interaction logic, no evaluation under motion.
Real-Time Pill Identification for Medicine Inspection Systems S. Kavitha, R. Madhumathy · 2025 Discover Electronics (Springer), 2:80	YOLOv5s plus a Deep Text Spotter and character-level RNN with coordinate encoding; trained to generalise from single- to multi-object scenes.	NLM Pill Dataset plus real-world pill images.	Accuracy plateaus near 98%; coordinate encoding reduces OCR error on partial imprints. We adopt its single-to-multi-object training strategy.	Industrial inspection: controlled lighting, fixed camera distance. Pills, not packaging. No patient-facing interface.
Fast and Accurate Medication Identification N. Larios Delgado et al. · 2019 npj Digital Medicine, 2:10	Deep CNN pipeline for prescription-pill identification from ordinary mobile photographs, framed around medication reconciliation.	NIH NLM Pill Image Recognition Challenge.	94% top-5 accuracy against the 83.3% challenge winner, and frames recognition as error reduction rather than a vision benchmark.	Cloud-scale model; pills not packaging; single still image, no spatial anchoring, no alerting layer.

Accuracy on this strand is effectively solved for closed sets. The open problems are packaging rather than pills, multiple objects per frame, and motion.

Strand C — Drug–drug interaction prediction

Paper / Authors / Venue	Methodology	Dataset & tools	Major findings and advantages	Limitations
DeepDDI: Deep Learning Improves Prediction of Drug–Drug and Drug–Food Interactions J. Y. Ryu, H. U. Kim, S. Y. Lee · 2018 PNAS, 115(18):E4304–E4311	Structural similarity profiles of a drug pair feed a deep network that predicts interaction type, using only names and structures.	DrugBank; drug and food-constituent structures.	Predicts typed interactions, not a yes/no flag, from inputs a scanner can obtain. Drug–food coverage maps onto our timing advice.	Pairwise and offline at server scale. No vision input, no latency budget, no ranking of alerts by urgency for a lay user.
DDIMDL: A Multimodal Deep Learning Framework for Predicting DDI Events Y. Deng et al. · 2020 Bioinformatics	Fuses several feature views — targets, enzymes, pathways, substructures — in a joint DNN over 65 interaction event types.	DrugBank-derived set, 65 event classes.	Event-type granularity is learnable and multi-view fusion beats any single view. Sets the bar our compressed model must approach.	Depends on curated pharmacological features unavailable at scan time. Heavyweight; no deployment or compression study.
SSI-DDI: Substructure–Substructure Interactions for DDI Prediction A. K. Nyamabo, H. Yu, J.-Y. Shi · 2021 Briefings in Bioinformatics, 22(6):bbab133	Graph attention network over molecular graphs; recasts a pair interaction as interactions between learned substructures.	DrugBank molecular graphs.	Attention weights localise which substructures drive the interaction — an explanation we can surface instead of an opaque warning.	Cold-start weakness on unseen drugs. No clinical interface; not pruned or quantised for mobile inference.
AutoDDI: DDI Prediction with an Automated Graph Neural Network J. Gao et al. · 2024 IEEE J. Biomed. Health Inform., 28(3):1773–1784	Neural architecture search over GNN design space, selecting message passing, aggregation and readout for the DDI task.	DrugBank, TWOSIDES.	Searched architectures beat hand-designed GNNs and remove manual tuning — architecture choice, not just data, still matters.	High search cost, and accuracy is the sole objective: no memory, latency or energy constraint, so nothing transfers to a handset.

Every model on this strand assumes drug identities are already known and that compute is unconstrained. None has been placed behind a camera or inside a real-time loop.

Strand D — Mobile AR tracking and registration

Paper / Authors / Venue	Methodology	Dataset & tools	Major findings and advantages	Limitations
Jaguar: Low Latency Mobile AR with Flexible Tracking W. Zhang, B. Han, P. Hui · 2018 ACM Multimedia (MM)	ARCore client gives markerless 6-DoF pose and size estimation; recognition is offloaded to an edge GPU server with an optimised retrieval pipeline.	ARCore; edge GPU server; retrieval benchmark.	End-to-end latency near the inter-frame interval, so recognition runs continuously without the user holding still — our target quality.	Requires an edge server, breaking privacy and offline use for medical data. Generic objects; no domain semantics or safety reasoning.
Real-Time Motion Tracking for Mobile AR/VR Using Adaptive Visual-Inertial Fusion W. Fang et al. · 2017 Sensors, 17(5):1037	Monocular visual-inertial SLAM with an adaptive filter that trades jitter against latency for smooth 6-DoF tracking.	Monocular camera + IMU; ground-truth trajectories.	Quantifies the jitter-versus-latency trade-off that decides whether an anchored label looks stable — basis for our jitter metric.	Device pose only, no object-level semantics, and drifts in low-texture scenes such as a plain shelf or countertop.
In-Depth Review of AR: Tracking, Development Tools, Displays, Collaborative AR and Security T. A. Syed et al. · 2023 Sensors, 23(1):146	Systematic review of marker, markerless, SLAM and sensor-fusion tracking, AR SDKs, displays and AR security.	Literature across IEEE, ACM, Springer, MDPI.	Maps the registration trade-offs behind our SDK choice and flags AR privacy as under-addressed — a gap our on-device design answers.	Review only; no implementation or measurement. Healthcare-specific safety requirements are not treated.

This strand supplies the pose and stability machinery but carries no domain knowledge; the medication strands carry domain knowledge but ignore registration quality.

Techniques, datasets, metrics and results by strand

Strand	Representative works	Technique / algorithm	Datasets	Performance metrics & results	Advantages	Limitations
A **AR medication assistance**	Rupa 2022; Blusi 2019; Ingeson 2018; Tangtisanon 2016	OCR + NLP + drug-name lookup; ARCore rendering; BDI agent on HoloLens; template matching	In-house DNR DB; real prescriptions; custom label sets	Functional-factor analysis; questionnaire acceptability (n = 15). No standard recognition metric reported	Markerless and low cost on phones; spatially situated; acceptable to users	One product per session; no dosage, duplicate or interaction logic; headsets unaffordable
B **Package / pill recognition**	Zeng 2017; Han 2021; Liu 2020; Kavitha 2025; Larios Delgado 2019	Multi-CNN + distillation; induced two-stage DL; image–text embedding; YOLOv5s + text spotter	NIH NLM PIR Challenge; NLM Pill Dataset; FDA labels; blister DB	94% top-5 (vs 83.3% winner); blister F1 to 99.98% (baselines 86.5–96.2%); ~98% train/val	Very high closed-set accuracy; on-device footprint proven; text + appearance fusion	Pills more than packaging; one object per frame; static capture; no reasoning layer
C **DDI prediction**	Ryu 2018; Deng 2020; Nyamabo 2021; Gao 2024	Similarity profile + DNN; multi-view fusion; GAT over substructures; GNN architecture search	DrugBank; TWOSIDES; 65-class DDI event set	Typed DDI classification; AUROC / AUPRC / macro-F1 on held-out pairs; NAS beats hand-designed GNNs	Typed and explainable predictions; best case needs only name + structure	Assumes drug IDs already known; server-scale compute; no latency or memory budget
D **AR tracking**	Zhang 2018; Fang 2017; Syed 2023	ARCore 6-DoF + edge-offloaded retrieval; adaptive visual-inertial fusion; SDK review	ARCore sessions; camera + IMU with ground truth	End-to-end latency near the inter-frame interval; jitter and drift measured against ground truth	Continuous recognition without holding still; quantified stability	Edge server breaks privacy and offline use; pose only, no semantics; drift on low texture

Reported figures come from different datasets and protocols and are not directly comparable; they are listed to show the level each strand has reached, not to rank the papers.

SECTION 4 · COMPARATIVE ANALYSIS

Capability coverage: where the reviewed work stops

Capability	A · AR medication assistance	B · Package / pill recognition	C · DDI prediction	D · AR tracking & registration	Proposed MediLens AR
Recognises consumer packaging, not loose pills	Yes	Partial	No	No	Yes
Several products in one camera frame	No	Partial	Yes	Yes	Yes
Overlay spatially anchored to the object	Yes	No	No	Yes	Yes
Dosage and timing guidance	Partial	No	No	No	Yes
Duplicate active-ingredient detection	No	No	No	No	Yes
Interaction alerting with severity tiers	No	No	Yes	No	Yes
Runs fully on-device, no data egress	Partial	Yes	No	No	Yes
AR and human-factors evaluation	Partial	No	No	Yes	Yes

Two rows are empty across the entire literature: same-salt duplicate detection, and any system that combines multi-object packaging recognition with interaction reasoning inside a registered AR view. That intersection is the project.

SECTION 5 · PROBLEM STATEMENT

The problem in one sentence

A patient holding two or more medicine packages has no way to find out, in the moment and without typing, whether those products share an active ingredient, interact dangerously, or are due at the same time — because the necessary information is printed in fine text on separate boxes and must be cross-referenced by hand.

WHO IS AFFECTED

Older adults on five or more medicines, their family caregivers, patients managing chronic disease across several prescribers, low-vision and low-literacy users, and pharmacy counter staff verifying an unfamiliar combination.

WHY IT MUST BE SOLVED

Polypharmacy is common and rising, therapeutic duplication is easy to commit when brand names differ, and the consequences — bleeding risk, renal injury, hospitalisation — are severe and largely preventable.

WHY EXISTING TOOLS FAIL

Interaction checkers require the user to already know and correctly spell every drug name. AR medication apps identify one box and stop. Neither reasons about the scene the user is actually looking at.

WHAT MAKES IT HARD

Recognition, ingredient resolution and interaction reasoning must all complete inside an AR frame budget, on a mid-range handset, without sending a medication list — sensitive health data — off the device.

SECTION 6 · RESEARCH GAP

Six gaps, and how this project answers each

1 Single-object ceiling

AR medication systems recognise one package per session, so relationships between products cannot exist in the view.

Our response Multi-package detection and tracking, with interaction edges drawn between the physical boxes.

2 Duplication is unaddressed

No reviewed work resolves different brand names to a shared active ingredient, the most common therapeutic-duplication error.

Our response A composition index maps brand → salts, flagging overlap before the interaction check runs.

3 No real-time implementation

DDI models are evaluated offline at server scale; none reports latency, memory or energy on a handset.

Our response Pruned and INT8-quantised GNN with a measured per-frame budget on mid-range hardware.

4 Privacy and security ignored

Edge-offloaded AR pipelines transmit frames; a medication list is sensitive health data.

Our response Fully on-device inference and a local knowledge store — zero patient-data egress by architecture.

5 Limited, non-representative data

Datasets are US and Korea-centric pill images; Indian brand-dominant, multilingual strip packaging is absent.

Our response IndiMed-AR: a purpose-built annotated multi-pack dataset of Indian packaging in realistic conditions.

6 Accuracy-only evaluation

Papers report classification metrics but rarely registration jitter, legibility, comprehension time or cognitive load.

Our response Joint evaluation: recognition and DDI metrics plus AR stability and human-factors measures.

SECTION 7 · RESEARCH OBJECTIVES

Six objectives for the project

1 To analyse the limitations of existing methods

Systematically review 17 recent works across AR medication assistance, packaging recognition, DDI prediction and AR tracking, and derive a capability coverage matrix.

2 To build a representative dataset

Construct IndiMed-AR: roughly 5,000 annotated images of about 250 Indian medicine packs and strips, including multi-pack scenes under varied lighting, angle and occlusion.

3 To design the perception module

Detect and track several packages simultaneously, read on-pack text, and resolve brand names to active ingredients through a fuzzy composition index.

4 To develop on-device safety reasoning

Implement a compressed interaction model plus a duplicate-salt checker and schedule engine that run locally, with no patient data leaving the handset.

5 To implement the AR presentation layer

Render severity-coded interaction edges between physical packages, per-pack dose-clock rings and persistent shelf anchors, with stable registration.

6 To evaluate against existing approaches

Measure recognition, interaction, real-time and human-factors performance, benchmarked against single-object AR identification and conventional leaflet reading.

System architecture: camera frame to anchored overlay

INPUT · camera frame stream + IMU · ARCore session providing 6-DoF device pose, plane detection and persistent anchors

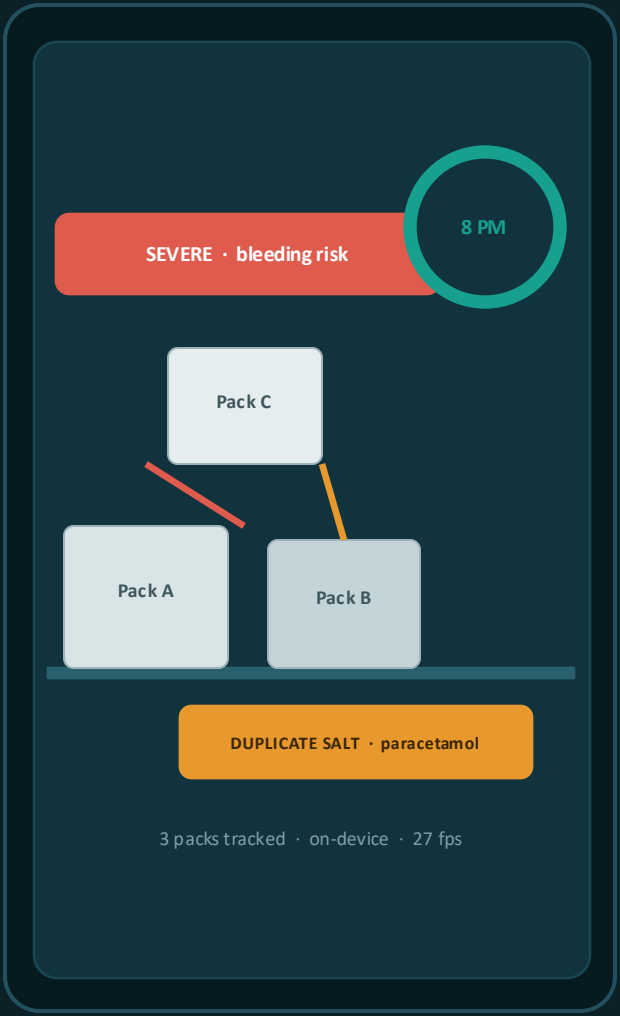
1 Perception	2 Resolution	3 Reasoning	4 Presentation
Multi-object package detector YOLOv8n / RT-DETR-lite, INT8	**Brand-name normalisation** edit distance + phonetic matching	**Duplicate-salt checker** set overlap across resolved packs	**Severity ranking & triage** severe / moderate / informational
Per-ROI text recognition lightweight CRNN + PaddleOCR-lite	**Brand → active ingredient** local composition index	**Interaction scorer** pruned GNN over cached DDI subgraph	**AR view manager** label placement, occlusion, anti-jitter
Pose association & tracking ByteTrack + Kalman smoothing	**Confidence gating** low-confidence packs marked unresolved	**Schedule & dose engine** timing, food rules, dose windows	**Multimodal output** anchored labels, edges, TTS, haptics

OUTPUT · spatially anchored AR layer · severity-coded interaction edges between packages · per-pack dose-clock rings · spoken and haptic alerts

Local knowledge store SQLite / Room — composition index, cached DDI subgraph, user regimen and shelf anchors. No frame, drug name or medication list is transmitted; the pipeline runs with the network disabled.

SECTION 8 · PROPOSED METHODOLOGY

The AR interaction layer



Illustrative concept view

1

Spatially anchored interaction graph

Severity-coded edges are drawn in 3D between the physical packages themselves, so the risk is attached to the objects the user is holding rather than listed in a separate panel.

2

Salt-level duplicate detection

Brand names are resolved to active ingredients before any interaction check, catching therapeutic duplication — two products, one ingredient — which no reviewed system attempts.

3

Dose-clock ring per package

A floating ring above each pack encodes the next dose window and its state (due, taken, overdue) together with food and spacing constraints, replacing a separate reminder list.

4

Cabinet memory via persistent anchors

The medicine shelf is saved as a persistent anchor set, so a new purchase can be checked against what the household already owns — including at the pharmacy counter.

SECTION 8 · PROPOSED METHODOLOGY

Datasets, preprocessing, and the training procedure

Dataset collection and selection

IndiMed-AR (self-collected)

~5,000 images of ~250 Indian OTC and prescription packs and strips. Single-pack and multi-pack scenes; varied lighting, angle, glare, partial occlusion and hand-held motion. Annotated with bounding boxes, brand, and active-ingredient set.

NIH NLM Pill Image Recognition Challenge / NLM Pill Dataset

Public benchmark used to pre-train the detector and to report comparable numbers against Zeng 2017, Kavitha 2025 and Larios Delgado 2019.

DrugBank (academic licence)

Interaction pairs and typed interaction events, plus ingredient and structure records used to build the local composition index.

TWOSIDES

Post-market adverse-event associations used as a secondary evaluation set for the interaction scorer.

Data preprocessing

- Perspective rectification of the detected pack face

CLAHE contrast equalisation and specular-glare suppression

Motion-blur, rotation, scale and colour-jitter augmentation

Synthetic multi-pack compositing to expand scene combinations

Text-region crops with character-level augmentation for the OCR head

Brand-string canonicalisation before composition lookup

Training and testing process

- Detector: pre-train on NLM, fine-tune on IndiMed-AR; 70 / 15 / 15 split held at the pack-identity level to prevent leakage

OCR head: trained on text-region crops with synthetic and real strings; CER and WER reported separately

Interaction scorer: GNN trained on DrugBank pairs with 5-fold cross-validation, then structurally pruned and INT8 quantised; accuracy re-measured after compression

Integration test: full pipeline profiled on-device over 10-minute continuous sessions, logging latency, frame rate, thermal state and battery drain

Ablations: with and without the composition index, and cloud FP32 versus on-device INT8

Dataset collection follows institutional ethics guidance: packaging is photographed with consent, and no prescription, patient identifier or personal regimen is recorded.

Evaluation metrics, software and hardware requirements

Perception

- mAP@0.5 and mAP@0.5:0.95 for multi-pack detection

Character and word error rate for on-pack OCR

Top-1 brand → active-ingredient resolution accuracy

Tracking ID switches per session

Reasoning

- AUROC, AUPRC and macro-F1 over interaction event types

Precision and recall for duplicate-salt detection

False-alarm rate at the operating threshold

Accuracy delta after pruning and INT8 quantisation

Real-time & system

- End-to-end latency per frame (ms)

Sustained frame rate on mid-range hardware

Registration jitter of anchored labels (mm)

Battery drain and thermal state over 10 minutes

Human factors

- Time to first correct safety decision

Error rate versus reading the printed leaflet

System Usability Scale (SUS)

NASA-TLX perceived workload

Software requirements

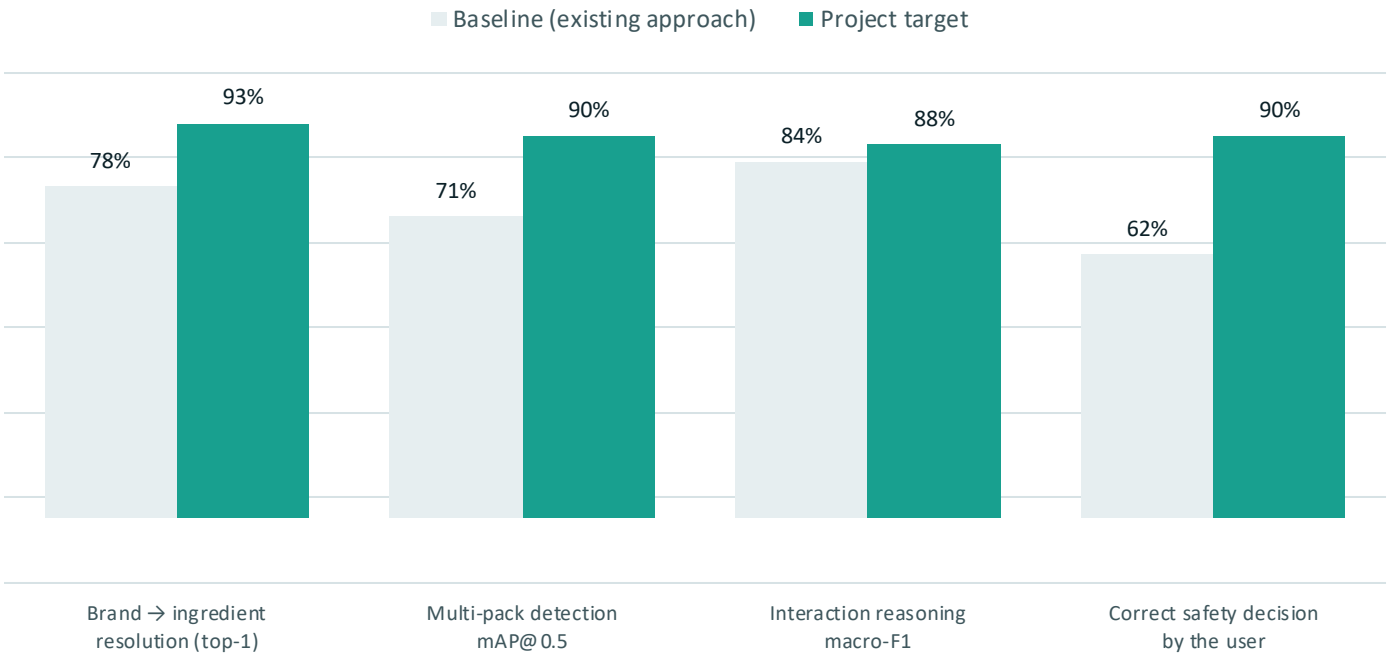
AR runtime Unity 2022 LTS · AR Foundation · ARCore XR Plugin
Inference TensorFlow Lite / ONNX Runtime Mobile with NNAPI and GPU delegates
Vision & OCR PaddleOCR-lite or ML Kit text recognition · OpenCV for rectification
Training PyTorch, PyTorch Geometric, Ultralytics; Roboflow / CVAT for annotation
Local store SQLite / Room with an encrypted regimen table

Hardware requirements

Target device ARCore-certified Android 12+ handset, 6 GB RAM, Snapdragon 7-series or better
Lower bound 4 GB RAM mid-range device tested as the degraded-performance floor
Training rig Single NVIDIA GPU with 12 GB or more VRAM (lab workstation or Colab Pro)
Capture Handheld phone cameras only — no depth sensor, headset or fiducial marker required

SECTION 9 · EXPECTED OUTCOMES

Target performance against current practice



≤ 120 ms

end-to-end latency per frame on the target handset

≥ 25 fps

sustained frame rate with three packs tracked

0 bytes

patient data leaving the device — pipeline runs offline

≥ 40%

reduction in time to identify a harmful combination

Also expected: improved scalability through a composition index that generalises to unseen brands without retraining; reduced computational cost from pruning and INT8 quantisation; better security by removing network dependence entirely; and a reusable annotated dataset of Indian medicine packaging released for future work.

Baseline values are drawn from the best comparable figures reported in the reviewed literature under different protocols; all figures on this slide are targets, not results.

SECTION 10 · WORK COMPLETED

Activities completed before Review 1

Topic finalisation

Complete

Scope fixed to markerless handheld AR for medication safety, with the AR interaction graph agreed as the core contribution.

Literature collection

Complete

62 candidate papers screened; 17 retained across four strands from IEEE, Springer, Elsevier, ACM, PNAS and PMC-indexed venues.

Literature survey

Complete

Each retained paper documented on methodology, dataset, findings, advantages and limitations.

Comparative analysis

Complete

Strand-level comparison table and an eight-capability coverage matrix built from the extracted limitations.

Problem identification

Complete

Problem statement drafted, affected populations characterised, and existing-approach failure modes catalogued.

Research gap identification

Complete

Six gaps derived from the coverage matrix and mapped one-to-one onto the project objectives.

Dataset selection & pilot capture

In progress

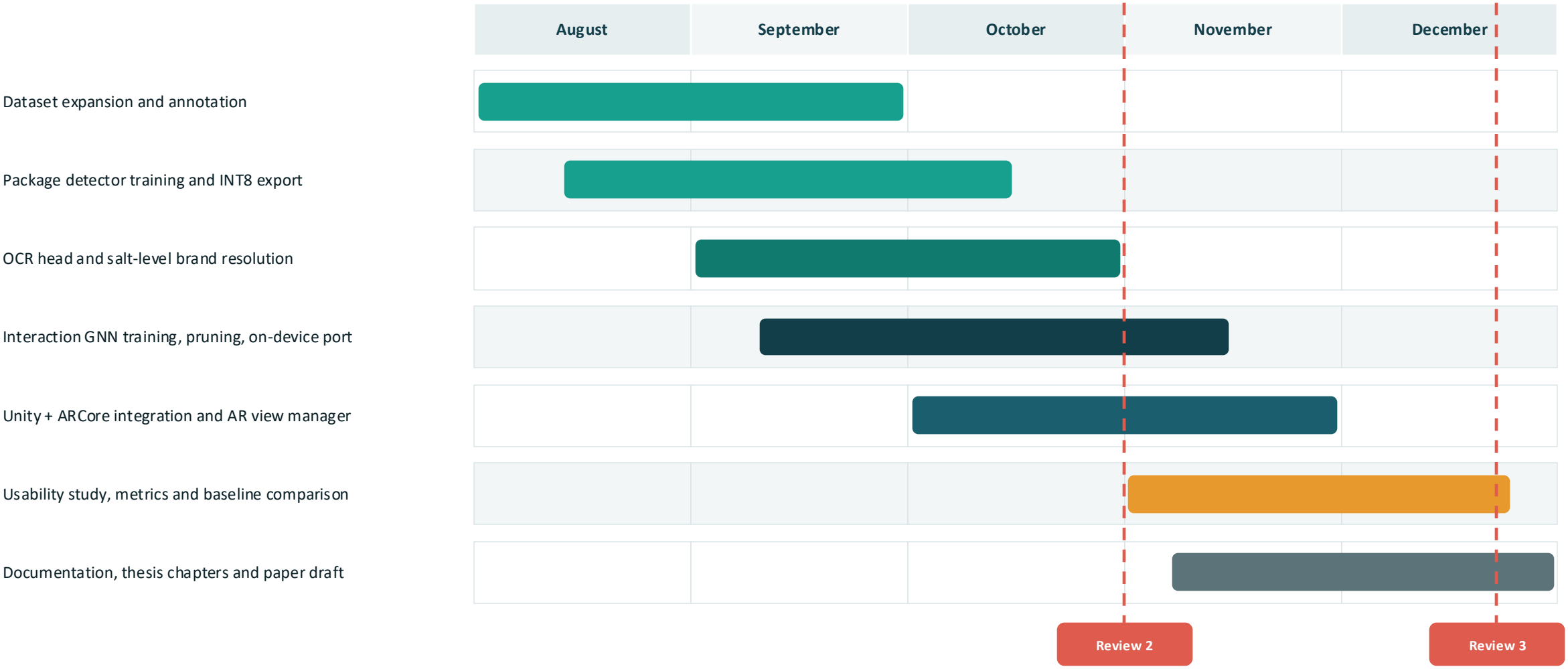
Public datasets secured; 620 pilot images of 48 packs captured and annotated to validate the labelling schema.

Initial architecture design

In progress

Four-stage architecture drafted; feasibility spike confirmed ARCore plus a TFLite INT8 detector at ~28 fps on the target device.

Planned activities through Review 2 and Review 3



Deliverable at Review 2: working multi-pack detection with salt-level resolution and a first anchored overlay. *Deliverable at Review 3: full pipeline on-device, evaluated results and comparison against baselines.*

SECTION 12 · REFERENCES

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SECTION 12 · REFERENCES

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Thank you

